Report: Privacy Shield — Tracker Blocker Browser Extension (Manifest V3)

Introduction

Websites frequently load third-party scripts for ads, analytics, and social widgets. These scripts can track users across multiple sites, reducing privacy and increasing unwanted profiling. The purpose of *Privacy Shield* is to provide a simple, standalone browser extension that blocks known trackers and helps users understand what was blocked through a clear UI and analytics.

Abstract

Privacy Shield is a privacy-focused browser extension built using Manifest V3 and vanilla HTML/CSS/JavaScript (no frameworks). It blocks known tracking domains using request-blocking rules, keeps a count of blocked trackers per tab (badge + popup), and maintains global analytics (totals, categories, daily trends, and top tracker domains). Users can control behavior using a whitelist/blacklist and can optionally log in using a simple JWT-based authentication stored locally.

Tools Used

- **JavaScript (ES6+)** — extension logic and UI scripting
- **Manifest V3** — modern extension structure (service worker background)
- **Declarative Net Request API** — blocking tracker requests via rulesets
- **Chrome/Firefox WebExtensions APIs** — tabs, storage, webNavigation, action badge
- **HTML/CSS** — popup UI, dashboard, options, auth pages (no React)
- **chrome.storage.local** — persistent settings and analytics storage
- **JWT (client-side)** — simple login/token handling for local authentication

Steps Involved in Building the Project

1. **Project Setup**: Created extension folder structure (background, rules, popup, dashboard, options, auth, utils).
2. **Manifest Configuration**: Added permissions and connected popup, options page, service worker, content script, and ruleset.
3. **Tracking List Creation**: Built a combined tracker domain list grouped into categories (analytics, advertising, social, fingerprinting, cryptomining).
4. **Request Blocking**: Implemented tracker blocking using Declarative Net Request rules and added dynamic rules for whitelist/blacklist.

5. **Badge + Popup Counter**: Counted blocked trackers per tab and displayed the count on the extension badge and popup UI.
6. **Analytics System**: Stored global stats (total blocked, daily blocked, category breakdown, top trackers) in chrome.storage.local.
7. **Dashboard Page**: Displayed analytics in a dashboard page with totals, category breakdown, daily trend, and top tracker list.
8. **User Controls**: Added options page for whitelist/blacklist management with import/export.
9. **JWT Authentication**: Implemented login/register UI and local JWT token storage for authenticated state.

Conclusion

This project delivers a complete, standalone tracker-blocking extension for Chromium browsers (Chrome/Brave/Edge) and WebExtensions-compatible environments. It blocks known trackers, provides a real-time badge counter, and offers an analytics dashboard to improve transparency. With whitelist/blacklist controls and a simple JWT login flow, the extension gives users both privacy protection and control in a lightweight, framework-free implementation.